Effects of Advanced Trauma Life Support® on Adult Trauma Patient Outcomes (ADVANCE TRAUMA): A Cluster Randomised Trial

Svenska Läkaresällskapets fonder, Projektansökan

Martin Gerdin Wärnberg

Institutionen för folkhälsovetenskap, ME Intensivvård och Thoraxoperation

Jag har läst och förstått ovanstående information

Prefekt/klinikchef eller motsvarande

Marie Klang Hasselberg

Medlemsförening

Ingen medlemsförening

Medlemsnummer

201,345

Prefekt/motsvarande är informerad

Ja

Tidigare tilldelad SLS projektanslag

Ja, FÄRRE än 5 gånger

Lämna en kort redogörelse över vad dessa medel gått till.

The previous project grant was important in funding the work of three PhD-students, Jonatan Attergrim, Kelvin Szolnoky and Johanna Berg. The grant has so far led to one published article and two articles under review. In our published article, we showed that opportunities for improvement were identified in 5.8% of trauma patients. Age, missing Glasgow Coma Scale, time to first CT, highest hospital care level and ISS were statistically significantly associated with opportunities for improvement. Building on these results, we developed machine learning models that predicts opportunities for improvement in trauma care and compared the performances of these models to those of commonly used audit filters. We found that both the logistic regression and XGBoost models outperformed audit filters in predicting opportunities for improvement among adult trauma patients and can potentially be used to improve systems for selecting patient cases for trauma peer review.

Läkarexamen

Ja

Legitimerad Läkare

Ja

Disputerad

Ja, för HÖGST 10 år sedan

Avdrag föräldraledighet eller annat

60

Docent

Ja

Ämnesområde/Andra projekt

Huvudsakligt ämnesområde

Övrigt

Specificera ämnesområde/n

Folkhälsovetenskap
Kirurgi
Medicin
Övrigt

Är du huvudman för annan ansökan

Nej

Aktuellt projekt

Typ av ansökan

Ny ansökan

Projektsammanfattning

Varje år dör 4.3 miljoner människor av trauma, alltså svår kroppskada som orsakas av olyckor eller våld. Ett bra primärt omhändertagande, alltså vård tidigt efter skadan, är avgörande för att rädda liv och minska sjuklighet. Flera kurser har tagits fram för att förbättra det primära omhändertagandet. Den mest kända kursen är Avanced Trauma Life Support®, och mer än en miljon läkare över hela världen har utbildats i ATLS ® sedan 1978.

Trots att ATLS används i så stor omfattning finns det ingen forskning av hög kvalitet som visar att ATLS ® förbättrar utfallen hos patienter. Detta är problematiskt eftersom det betyder att många länder, inklusive Sverige, spenderar miljontals kronor varje år på att utbilda läkare i ATLS ® utan att veta om det leder till förbättrad överlevnad eller minskad sjuklighet hos patienter. Vi vill därför studera om ATLS ® förbättrar överlevnad och minskar sjuklighet hos vuxna traumapatienter.

Vi genomför projektet i Indien eftersom ATLS ® inte är standard där än. Vårt projekt blir det första att utvärdera effekten av ATLS ® på utfall hos patienter genom storskalig forskning av hög kvalitet. Vårt projekt är viktigt oavsett om resultaten är positiva eller negativa. Om vi visar att ATLS ® är effektivt bör liknande kurser införas i större skala, men om det inte är så behövs nya sätt att träna det primära omhändertagandet av traumapatienter.

Projektbeskrivning

Purpose and aim

Each year, trauma - physical injury and the body's associated response - kills over four million people [1], is the top cause of burden of disease in young people aged 10-49 years [2], and costs trillions of dollars [3]. Advanced Trauma Life Support ® (ATLS ®) is the most established training programme aiming to help physicians in the initial management of trauma patients [4], but high-quality evidence from randomised trials showing that ATLS ® improves patient outcomes is missing [5–8]. We will conduct a cluster randomised trial to compare the effects of ATLS ® training with standard care on outcomes in adult trauma patients (ClinicalTrials.gov ID NCT06321419). The trial will be conducted in India and is funded by the Swedish Research Council (Dnr. 2023-03128) and the Laerdal Foundation (reg. no. 2023-0297), which covers including ten of the required 30 clusters. Here we apply for additional funding to recruit the required number of clusters.

State-of-the-art and preliminary results

There are three randomised controlled studies showing that ATLS ® improves knowledge and clinical skills [5], but systematic reviews have failed to identify any randomised or high-quality quasi-experimental trials indicating that ATLS ® improves patient outcomes [5–8]. Since 2020, we have conducted a series of systematic searches in collaboration with the Karolinska Institutet University Library to identify studies on the impact of ATLS ® training on patient outcomes, and our latest search was conducted on January 22, 2025. We have screened a total of 10,429 records and estimated a pooled effect of ATLS ® training on mortality from 12 heterogeneous (I^2 0.9) observational studies as a risk ratio of 0.76 (95% CI 0.57; 1.01) (Attached Figure 1) [9–20]. These studies indicate that ATLS ® may reduce mortality, but that the quality of the evidence is low.

We conducted a pilot cluster randomised controlled trial (ClinicalTrials.gov NCT05417243) between April 2022 and February 2023 to assess the feasibility of a full scale trial. We published the protocol for this pilot study [21] and the results are available as a preprint [22]. Our pilot study enrolled 375 patients from seven hospitals across India and shows that it is feasible to conduct the proposed trial with a high inclusion rate and low missingness in key variables. To involve patients and the public in the planning of this trial we conducted semi-structured interviews with trauma patients, caregivers, and community representatives. The aim of these interviews was to understand their views on the trial and important outcomes and they showed high acceptability of our research and emphasised the importance of better recovery before discharge and reduced disability. The interviews also highlighted return to work as an important outcome. The results are also available as a preprint [23].

Methods

This project was developed jointly by the parties participating in the Trauma life support training Effectiveness Research Network (www.tern.network), funded by the Swedish Research Council (Dnr. 2020-03779). We report our methods according to the Consolidated Standards Of Reporting Trials (CONSORT) extension for stepped-wedge cluster randomised controlled trials [24]. The trial is registered with ClinicalTrials.gov, number NCT06321419, and approved by the relevant regulatory and ethical review authorities.

<u>Design</u>

We will conduct a batched stepped-wedge cluster randomised controlled trial in India (Attached Figure 2). We conduct this trial in India because of 1) our established collaboration with Indian institutions and experience in conducting multicentre studies in this setting (www.titco.org), and 2) physicians in India are not routinely trained in ATLS ® or similar programmes. We will roll out ATLS ® to 30 clusters over six batches, so there will be five clusters in each batch. The clusters in each batch will be randomised to one of five implementation sequences, with one cluster randomised to each implementation sequence. All clusters will transition through three phases, first a standard care phase, then a transition phase during which the training is delivered, and finally an intervention phase, for a total of 13 months. The implementation sequence determines how long the phases of standard care and intervention are.

<u>Participants</u>

Our trial has eligibility criteria on three levels: hospitals, clusters and patient participants. Hospital will be the unit of randomisation. To be eligible, **hospitals** need to admit at least 400 trauma patients per year and provide surgical and orthopaedic emergency services around the clock. **Clusters** will be one or more units of physicians providing initial trauma care in the emergency department of included hospitals. These units must admit or refer/transfer for admission at least 12 patients with trauma per month. **Patient participants** will be adult trauma patients who are admitted at participating hospitals. We define a history of trauma using the International Classification of Diseases chapter 20.

Intervention and control treatment

The intervention will be **Advanced Trauma Life Support ® (ATLS ®)** [4] training. This is a proprietary 2.5 day course teaching a standardised approach to intial treatment and resuscitation, triage and interfacility transfers of trauma patients using the concepts of a primary and secondary survey. The programme was developed by the Committee of Trauma of the American College of Surgeons. The control will be **standard care**, meaning no formal trauma life support training. During the transition phase, we will train the physicians that initially resuscitate and provide trauma care during the first hour after patient arrival at the emergency department. These physicians can be casualty medical officers, surgical residents, or emergency medicine residents, depending on the setup at each participating centre.

Outcomes

We chose outcomes that we judged as clinically important and that patients, their caregivers and community representatives perceived as important in our interviews with them. The **primary outcome is in-hospital mortality within 30 days of arrival at the emergency department**. The **secondary outcomes** are mortality at additional time points, quality of life (EQ5D5L), disability (WHODAS 2.0), and return to work. These outcomes will be measured up to three months after arrival at the emergency department.

Data analysis and statistics

We will conduct all analyses by modified intention to treat. All analyses will consider the clustered nature of the stepped-wedge design and we will adjust for temporal confounding. For the primary outcome, we will use mixed effects binomial regression with a logit-link to estimate an odds ratio, and a binomial model with identity link to estimate the risk difference [25]. Clusters and observations within clusters will be considered exposed to the intervention after the date at which the cluster was scheduled to transition. We will use a two-sided significance level of 5% and estimate 95% confidence intervals.

Power analysis

To detect a reduction in all-cause mortality within 30 days from 20% under standard care to 15% after ATLS ® training as significant at the 5% level, with ~90% power across different combinations of cluster autocorrelations and intra-cluster correlations [26], we need a total sample size of 4320 patients across 30 clusters (Attached Figure 3). This mortality effect is a conservative estimate equalling a risk ratio of 0.75, which would be both clinically important and consistent with our pilot study and updated systematic review. We assume that each cluster will contribute approximately 12 observations per month to the analysis, based on our previous work.

Work plan

The funding from this grant will be used to recruit clusters in batch 3 and 4, which will be completed and reported within the 2-3 year time frame of this grant. This trial is a collaborative effort between researchers at Karolinska Institutet, the George Institute for Global Health in India, and the University of Birmingham in the UK. It will run for a total of five years, starting in February 2025. We have obtained ethical and regulatory approvals. Funding for batch 1 and 2 has already been secured through grants from the Swedish Research Council and Laerdal Foundation.

Importance

Our project will result in the first robust evidence on the effects of ATLS ® on patient outcomes since the programme was introduced in 1978. We move the research frontier beyond the prevailing paradigm of observational research [9–20], and set a new standard for research on trauma system interventions. We fill knowledge gaps because systematic reviews have repeatedly called for high-quality trials in settings where

ATLS ® is not routinely taught [5–7]. If our results are positive, and ATLS ® improves outcomes over standard care, then this programme should be promoted and ways to increase its uptake should be explored. If the results are negative, then new ways to perform trauma life support training are needed.

Bilagor

visual-abstract-and-figures.pdf
Visual abstract and figures

Referenser

References
1. GBD 2019 Diseases and Injuries Collaborators. Injuries—level 1 cause. The Lancet 396, (2020).
2. GBD 2019 Diseases and Injuries Collaborators. Global burden of 369 diseases and injuries in 204 countries and territories, 1990–2019: A systematic analysis for the global burden of disease study 2019. The Lancet 396, 1204–1222 (2020).
3. Peterson, C. et al. Economic Cost of Injury — United States, 2019. MMWR. Morbidity and Mortality Weekly Report 70, 1655–1659 (2021).
4. Committee on Trauma. Advanced trauma life support® student course manual. (American College of Surgeons, 2018).
5. Mohammad, A. et al. Educational and clinical impact of advanced trauma life support (ATLS) courses: A systematic review. World J. Surg. 38, 322–329 (2013).
6. Jayaraman, S. et al. Advanced trauma life support training for hospital staff. Cochrane Database Syst Rev (2014).
7. Jin, J. et al. Effectiveness of quality improvement processes, interventions, and structure in trauma systems in low- and middle-income countries: A systematic review and meta-analysis. World J. Surg. 45, 1982–1998 (2021).
8. Putra, A. B. et al. Impact of Advanced Trauma Life Support Training for Improving Mortality Outcome: A Systematic Review and Meta-analysis. The New Ropanasury Journal of Surgery 8, (2023).
9. Vestrup, J. A. et al. Impact of advanced trauma life support training on early trauma management. Am J Surg 155, 704–707 (1988).
10. Ariyanayagam, D. C. et al. The impact of the ATLS course on traffic accident mortality in trinidad and tobago. West Indian Med J 41, 72–74 (1992).
11. Ali, J. et al. Trauma outcome improves following the advanced trauma life support program in a developing country. J Trauma 34, 890–899 (1993).
12. Olson, C. J. et al. Influence of trauma system implementation on process of care delivered to seriously injured patients in rural trauma centers. Surgery 130, 273–279 (2001).
13. Olden, G. D. J. van et al. Clinical impact of advanced trauma life support. Am J Emerg Med 22, 522–525 (2004).
14. Wang, P. et al. Comparison of severe trauma care effect before and after advanced trauma life support training. Chin J Traumatol 13, 341–344 (2010).
15. Drimousis, P. G. et al. Advanced trauma life support certified physicians in a non trauma system setting: Is it enough? Resuscitation 82, 180–184 (2011).
16. Hashmi, Z. G. et al. Hospital-based trauma quality improvement initiatives. J Trauma 75, 60–68 (2013).
17. Petroze, R. T. et al. Can focused trauma education initiatives reduce mortality or improve resource utilization in a low-resource setting? World J. Surg. 39, 926–933 (2014).
18. Bellanova, G. et al. How formative courses about damage control surgery and non-operative management improved outcome and survival in unstable politrauma patients in a mountain trauma center. Annali italiani di chirurgia 87, 68–74 (2016).
19. Magnone, S. et al. Impact of ATLS Guidelines and Trauma Team Introduction an 24 Hours Mortality in Severe Trauma in a Busy Italian Metropolitan Hospital: A Case Control Study. Ulus Travma Derg (2015).
20. Kamau, C. et al. Impact of advanced trauma life support training on 30-day mortality in severely injured patients at a Kenyan tertiary center: A retrospective matched case-control study. International Journal of Emergency Medicine 17, 153 (2024).
21. Gerdin Wärnberg, M. et al. A pilot multicentre cluster randomised trial to compare the effect of trauma life support training programmes on patient and provider outcomes. BMJ Open 12, e057504 (2022).
22. Gerdin Wärnberg, M. et al. Effects of Trauma Life Support Training on Patient Outcomes: A Pilot Cluster Randomised Trial. medRxiv (2024).
23. David, S. et al. Patient-reported outcomes relevant to post-discharge trauma patients in urban India. medRxiv (2024).
24. Hemming, K. et al. Reporting of stepped wedge cluster randomised trials: Extension of the CONSORT

2010 statement with explanation and elaboration. BMJ k1614 (2018).
25. Li, F. et al. Mixed-effects models for the design and analysis of stepped wedge cluster randomized trials: An overview. Stat Methods Med Res 30, 612–639 (2020).
26. Hemming, K. et al. A tutorial on sample size calculation for multiple-period cluster randomized parallel, cross-over and stepped-wedge trials using the shiny CRT calculator. Int J Epidemiol 49, 979–995 (2020).

Klinisk relevans eller förankring

Advanced Trauma Life Support® (ATLS®) is the most widely used training programme for physicians managing trauma patients. Since the first course in 1978, over one million physicians in more than 80 countries have been trained. In many high-income countries, ATLS® is nearly mandatory for trauma care physicians. In Sweden, for example, it is recommended as part of residency training for surgery, neurosurgery, orthopaedics, and anaesthesiology.

Despite its widespread use, several concerns exist:
•ATLS® is costly for individuals and health systems in both high- and low-income countries;
•it has been criticised for infrequent updates and lack of contextual adaptation;
•some ATLS® concepts, such as its categorisation of haemorrhagic shock, lack clinical and theoretical support; and
•there is limited evidence on the effectiveness of individual components.

These concerns, along with the absence of high-quality evidence showing ATLS® improves patient outcomes, have led to repeated calls for large-scale trials. However, none have been conducted. This project is the first to challenge a longstanding trauma care dogma, with direct clinical relevance for training trauma care physicians. A positive result reinforces ATLS® use and should guide its optimisation. A negative result calls for new approaches to trauma life support training.

Beskriv din roll i projektet

I am the principal investigator of this project and am responsible for the overall design and conduct of the trial. I have conducted multicentre interventional and observational studies in India for over 10 years. I was the lead applicant for the grants from the Swedish Research Council and Laerdal Foundation that have been awarded to fund this project so far.

As the principal investigator, I am part of all trial committees, which have all been established. My roles in these committees are:
•I am a member of the Trial Team that runs the day to day operations of the trial;
•I chair the Trial Management Group that is responsible for the overall management of the trial and includes 17 members from institutions in Sweden, India and the UK; and
•I participate in the meetings of the Joint Trial Steering and Data Monitoring Committee, which is the trial's oversight body and is chaired by an external member and includes two external clinical experts, an external biostatistician, and a lay representative.

My time in this project is 30% of full time, and is funded by the Swedish Research Council. Lovisa Strömmer and Li Felländer-Tsai are both members of the Trial Management Group and contribute with clinical expertise in trauma surgery and training interventions.

Medsökande

Lovisa Strömmer

Capio AB

Li Felländer-Tsai

Ortopedi

Behov av brukarmedverkan (inkl. motivering)

We involved patients and the public in the planning of this trial by conducting semi-structured interviews with trauma patients, caregivers, and community representatives. The aim of these interviews was to understand their views on the trial and what outcomes they considered important. In the ongoing trial we have a lay

representative in the Joint Trial Steering and Data Monitoring Committee, ensuring that the public is represented in the trial's oversight body.

Anser du att forskningen i denna ansökan är av translationell karaktär?

Nej

Fem mest relevanta publikationer

first authorships

1

articles as first author

last authorships

3

articles as last author

1. Measuring socioeconomic outcomes in trauma patients up to one year post-discharge: A systematic review and meta-analysis.
   David SD, Aroke A, Roy N, Solomon H, Lundborg CS, Gerdin Wärnberg M.
   Injury 2022:53(2):272-285.
   Web of Science® citations 14
   PMID 34706829 10.1016/j.injury.2021.10.012
2. 'Coming home does not mean that the injury has gone'—exploring the lived experience of socioeconomic and quality of life outcomes in post-discharge trauma patients in urban India.
   David S, Roy N, Stålsby Lundborg C, Gerdin Wärnberg M, Solomon H.
   Global Public Health 2022:6 February
   Link
3. Measuring post-discharge socioeconomic and quality of life outcomes in trauma patients: a scoping review.
   David SD, Roy N, Solomon H, Lundborg CS, Wärnberg MG.
   Journal of patient-reported outcomes 2021:5(1):68.
   PMID 34370128 10.1186/s41687-021-00346-6
4. A pilot multicentre cluster randomised trial to compare the effect of trauma life support training programmes on patient and provider outcomes.
   Gerdin Wärnberg M, Berg J, Bhandarkar P, Chatterjee A, Chatterjee S, Chintamani C, Felländer-Tsai L, Gadgil A, Ghag G, Hasselberg M et al;
   Trauma life support training Effectiveness Research Network (TERN) collaborators.
   BMJ Open 2022:12(4):e057504.
   Web of Science® citations 8
   PMID 35437251 10.1136/bmjopen-2021-057504
5. Mortality following surgery for trauma in an Indian trauma cohort.
   Mansourati M, Kumar V, Khajanchi M, Saha M, Dharap S, Seger R, Warnberg M.
   Br J Surg 2018:105(10):1274-1282.
   Web of Science® citations 7
   PMID 29665031 10.1002/bjs.10862

Etisk prövning

Är gjord

Ethics applications

Effekter av träning i traumaomhändertagande på utfall hos vuxna traumapatienter: en klusterrandomiserad studie

Application no
2024-07547-01
Authority
Etikprövningsmyndigheten
Principal investigator
Martin Gerdin Wärnberg
Organisation
Karolinska Institutet
Decision
Godkänd
Decision date
2024-12-09

Väntar på beslut från etikprövningsnämnden

Nej

Tidigare erhållna bidrag från SLS eller andra bidragsgivare till detta eller närliggande projekt

Granted Karolinska Institutet
Community perceptions of outcomes in trauma life support training effectiveness research

Granted sum
122100 SEK
Date of decision
-0001-11-30
Year
2022
Principal investigator
Martin Gerdin Wärnberg
Granted Vetenskapsrådet
Does trauma life support training improve patient outcomes? Establishing a network of Swedish and Indian researchers

Granted sum
795000 SEK
Year
2021
Principal investigator
Martin Gerdin Wärnberg
Co-applicant
Marie Hasselberg
Co-applicant
Lovisa Strömmer
Co-applicant
Li Felländer-Tsai
Co-applicant
Anita Gadgil
Co-applicant
Nobhojit Roy
Co-applicant
Rajdeep Singh
Granted Laerdal Foundation
A Pilot Multicentre Cluster Randomized Trial to Compare The Effect of Trauma Life Support Training Programs on Patient and Provider Outcomes

Granted sum
220000 SEK
Date of decision
-0001-11-30
Year

2021
Principal investigator
Martin Gerdin Wärnberg
Granted Vetenskapsrådet (2023-03128)
Effects of Advanced Trauma Life Support® Training Compared to Standard Care on Adult Trauma Patient Outcomes: A Cluster Randomised Trial

Granted sum
7550000 SEK
Date of decision
2023-10-31
Year
2029
Recipient
Martin Gerdin Wärnberg
Duration of the grant
2024-01-01
Principal investigator
Martin Gerdin Wärnberg
Granted Laerdal Foundation (2023-0297)
Effects of Advanced Trauma Life Support® Training Compared to Standard Care on Adult Trauma Patient Outcomes: Initiating A Cluster Randomised Trial

Granted sum
400000 SEK
Date of decision
2023-12-19
Year
2024
Recipient
Martin Gerdin Wärnberg
Duration of the grant
2024-01-01
Principal investigator
Martin Gerdin Wärnberg

Redovisning av tidigare erhållna anslag från SLS (publikationer)

1. Albaaj H, Attergrim J, Strömmer L, Brattström O, Jacobsson M, Wihlke G, et al. Patient and process factors associated with opportunities for improvement in trauma care: a registry-based study. Scand J Trauma Resusc Emerg Med. 2023 Nov 27;31(1):87.
2. Attergrim J, Szolnoky K, Strömmer L, Brattström O, Whilke G, Jacobsson M, et al. Predicting Opportunities for Improvement in Trauma Care: A Registry-Based Cohort Study [Internet]. medRxiv; 2024 [cited 2025 Feb 1]. p. 2023.01.19.23284654. Available from: https://www.medrxiv.org/content/10.1101/2023.01.19.23284654v4
3. Berg J, David S, Bakhshi GD, Basak D, Chatterjee S, Soni KD, et al. Effects of trauma quality improvement program implementation on mortality: A nonrandomized controlled trial. medRxiv. 2024 Sep 25;2024.05.27.24307748.
4. Kapitan E, Berg J, David S, Mohan LN, Felländer-Tsai L, Chatterjee S, et al. The effect of trauma quality improvement programme implementation on quality of life among trauma patients in urban India. medRxiv. 2024 Nov 26;2024.09.28.24314529.

Budget år 1

	Data collection and administering the intervention in the five hospitals in batch 3 by the George Institute for Global Health in India.

Söker nu totalt från SLS år 1

	Sum
Söker totalt år 1	500000
Sum	**500,000**

Budget år 2

	Data collection and administering the intervention in the five hospitals in batch 4 by the George Institute for Global Health in India.

Söker nu totalt från SLS år 2

	Sum
Söker totalt år 2	500000
Sum	**500,000**

Utbetalningsuppgifter

The account / cost center is set

Härmed försäkrar jag på heder och samvete att uppgifterna i min ansökan är korrekta

vald

Härmed försäkrar jag på heder och samvete att uppgifterna i ovanstående redovisning är korrekta

vald